

● RS002 Week 11 — English Worksheet

Topic: Functions — Reusing Code · **Course:** Pokédex App · **Time:** about 45 minutes

This week you learn `def` properly. A **function** is a named block of code you can run again and again. You pass it **parameters** (the values it works on) and you get back a result with `return`. Functions are how you stop copying and pasting the same lines.

1 · Recall

From memory:

Write a helper that takes a dictionary `p` and prints `p['name']` and `p['type']`.

2 · Predict 🧠

Read each block. Before you run it, write what you think will happen.

```
def greet_trainer(name):  
    print(f"안녕하세요, {name} 트레이너님!")  
  
greet_trainer("지우")  
greet_trainer("이슬")
```

What two lines print?


```
def add_stats(p):  
    return p["hp"] + p["attack"]  
  
pikachu = {"hp": 35, "attack": 55}  
total = add_stats(pikachu)  
print(total)
```

return hands a value back. What number does **print(total)** show?


```
def find(pokedex, target):  
    for p in pokedex:  
        if p["name"] == target:  
            return p  
    return None  
  
dex = [{"name": "피카츄"}, {"name": "파이리"}]  
print(find(dex, "리자몽"))
```

리자몽 is not in the dex. What does the function return, and what prints?

3 · Spot the Bug 🐛

Read what each block is **supposed** to do, fix it, then explain the fix.

Bug A — This function should be **called** so the greeting prints. Right now it is only defined, so nothing happens.

```
def greet_trainer(name):  
    print(f"안녕하세요, {name} 트레이너님!")
```

Hint: define is not the same as call — add a line that runs it.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — This should **return** the total so the caller can use it. Right now it prints inside and returns nothing, so `total` is `None`.

```
def add_stats(p):  
    print(p["hp"] + p["attack"])  
  
pikachu = {"hp": 35, "attack": 55}  
total = add_stats(pikachu)  
print(f"총합: {total}")
```

Hint: use `return` instead of `print` inside the function.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — `find` should give back `None` when nothing matches, so the caller can react. Right now it returns nothing after the loop and the `if result` check breaks.

```
def find(pokedex, target):
    for p in pokedex:
        if p["name"] == target:
            return p

dex = [{"name": "피카츄"}]
result = find(dex, "리자몽")
if result != None:
    print("찾음")
else:
    print("못 찾음")
```

Hint: add `return None` after the loop.

Write the fixed code:

Why was it wrong? Why does your fix work?

4 · Modify It

Start from these working functions:

```
def show_pokemon(pokemon):
    print(f"\n=== {pokemon['name']} ===")
    print(f"타입: {pokemon['type']}")
    print(f"체력: {pokemon['hp']}")
    print(f"공격력: {pokemon['attack']}")

def find_pokemon(pokedex, search_name):
    for pokemon in pokedex:
        if pokemon["name"] == search_name:
            return pokemon
    return None

pokedex = [
    {"name": "피카츄", "type": "전기", "hp": 35, "attack": 55},
    {"name": "이상해씨", "type": "풀", "hp": 45, "attack": 49},
]

result = find_pokemon(pokedex, "피카츄")
if result != None:
    show_pokemon(result)
else:
    print("도감에 없습니다.")
```

Make these changes one at a time and run after each:

1. Call `find_pokemon` with a name that is **not** in the dex and watch the `else` branch run.
2. Add a line inside `show_pokemon` that prints a power total (`hp + attack`).
3. Write a tiny new function `count_dex(pokedex)` that **returns** how many Pokémon there are, then print it.

Write your changed / added lines here:

5 · Make It 📷

Take your Week 10 app and pull its **two core actions** — showing a Pokémon and searching the dex — into **functions**. `find_pokemon` should `return` the match (or `None`); `show_pokemon` should print one Pokémon's card.

When it works, send a **photo or video** on KakaoTalk, then explain what you did. Use these starters — write 4 to 6 sentences.

First, I wrote a function called ... that ...

I passed it the parameter(s) ...

My `find_pokemon` returned ... when nothing matched.

A function with `return` is different from one without because ...

The trickiest part was ...

Functions make my code cleaner because ...

6 · Record Your Walkthrough 🎥

Film it running. Teach it like the viewer is new. Try to use: **function**, **def**, **parameter**, **return**, **None**.

1. Run a search that finds a Pokémon, then one that does not.
2. Read your `find_pokemon` out loud and point to the `return`.
3. Explain the difference between a function that returns and one that just prints.
4. Show how calling a function twice saves you from copying code.

Plan what you will say here first:

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.